Test results — Aletheia Evidence Bundle v0.1

Aletheia Evidence Bundle v0.1 bundles one local file as evidence and verifies the bundle's integrity. Standard library only. It proves preservation/hash integrity only — never truth of contents. All commands below were run from a **clean extraction** of `aletheia_evidence_bundle_v01.zip`.

Authoritative freeze gate — each command passes individually (clean extraction)

The freeze gate is the explicit list of individual commands (see RELEASE_COMMANDS.md) plus the static manifest check. No wrapper orchestrator is in
the gate.

Command	Result
`python3 -m compileall -q .`	PASS
`python3 -m unittest discover -s tests -v`	PASS — 55 tests, ~0.4s
`python3 -O -m unittest discover -s tests -v`	PASS — 55 tests
`python3 selfcheck.py`	PASS — 8/8 gates
`python3 -O selfcheck.py`	PASS — 8/8 gates
`python3 -W error::ResourceWarning -m unittest discover -s tests`	PASS
`sha256sum -c aletheia_evidence_bundle_v01.sha256`	OK
`python3 verify_release_manifest.py`	PASS — 7/7 static checks, **no subprocesses**

Required behaviours (all covered)

#	Required test	Where
1	Valid case ID accepted	`test_validation.py::test_1_valid_case_id_accepte`

#	Required test	Where
2	Unsafe case ID refused	`::test_2_unsafe_case_id_refused`
3	Valid evidence ID accepted	`::test_3_valid_evidence_id_accepted`
4	Unsafe evidence ID refused	`::test_4_unsafe_evidence_id_refused`
5	Missing source file refused	`::test_5_missing_source_refused`
6	Directory source refused	`::test_6_directory_source_refused`
7	Symlink source refused	`::test_7_symlink_source_refused`
8	Oversized source refused	`::test_8_oversized_source_refused`
9	Valid file bundles successfully	`test_build_bundle.py::test_9_valid_file_bundles_`
10	Bundle copies file into evidence/evidence_001.bin	`::test_10_bundle_copies_file_into_evidence_bin`
11	Manifest is canonical JSON	`::test_11_manifest_is_canonical_json`
12	Checksums file written	`::test_12_checksums_file_written`
13	Verify valid bundle succeeds	`test_verify_bundle.py::test_13_verify_valid_bund`
14	Mutated evidence file fails verify	`::test_14_mutated_evidence_file_fails`
15	Mutated manifest fails verify	`::test_15_mutated_manifest_fails`
16	Missing evidence file fails verify	`::test_16_missing_evidence_file_fails`
17	Missing manifest fails verify	`::test_17_missing_manifest_fails`
18	Missing checksums fails verify	`::test_18_missing_checksums_fails`
19	Duplicate checksum key fails	`test_checksums.py::test_19_duplicate_key_fails`

#	Required test	Where
20	Unknown checksum key fails	`::test_20_unknown_key_fails`
21	Malformed checksum digest fails	`::test_21_malformed_digest_fails`
22	Unknown extra file in bundle fails	`test_verify_bundle.py::test_22_unknown_extra_fil`
23	Manifest source hash mismatch fails	`::test_23_manifest_source_hash_mismatch_fails`
24	Manifest bundled hash mismatch fails	`::test_24_manifest_bundled_hash_mismatch_fails`
25	Non-canonical manifest fails	`::test_25_non_canonical_manifest_fails`
26	VERIFIED_PRESENT does not claim truth	`::test_26_verified_present_is_not_a_truth_claim` `test_no_bare_asserts.py::TestHonesty`
27	Selfcheck passes	`test_selfcheck.py::test_27_selfcheck_passes_in_p`
28	No bare runtime asserts	`test_no_bare_asserts.py::test_28_no_bare_asserts`
29	`python -O` unchanged behaviour	full suite + selfcheck pass identically under `-O`
30	manifest verifier passes, no subprocesses	`test_no_bare_asserts.py::test_30_manifest_verifi`

Additional coverage: checksums round-trip, missing-key and malformed-line and uppercase-digest rejection; manifest field/shape/size/sha/status validation; deterministic manifest under a fixed timestamp; non-empty output dir refused; extra file inside `evidence/` refused; missing bundle dir fails.

Manual smoke test (NOT part of the freeze gate)

```
printf "hello evidence\n" > /tmp/evidence.txt
python3 aletheia.py bundle-file --file /tmp/evidence.txt \
  --case-id case_001 --evidence-id evidence_001 --out
/tmp/aletheia_bundle
python3 aletheia.py verify --bundle /tmp/aletheia_bundle
```

Confirmed: prints `Aletheia evidence bundle verified` and `evidence_status = VERIFIED_PRESENT`; the bundle contains `evidence_manifest.json`, `evidence_checksums.txt`, and `evidence/evidence_001.bin`; mutating the evidence file, the manifest, the checksums, or adding an unknown file makes verify fail with the correct `FAILED_*` status.

Scope

One local file → deterministic evidence bundle → hash/checksum verification. Nothing else. Not connected to the workflow. No new features beyond the spec.